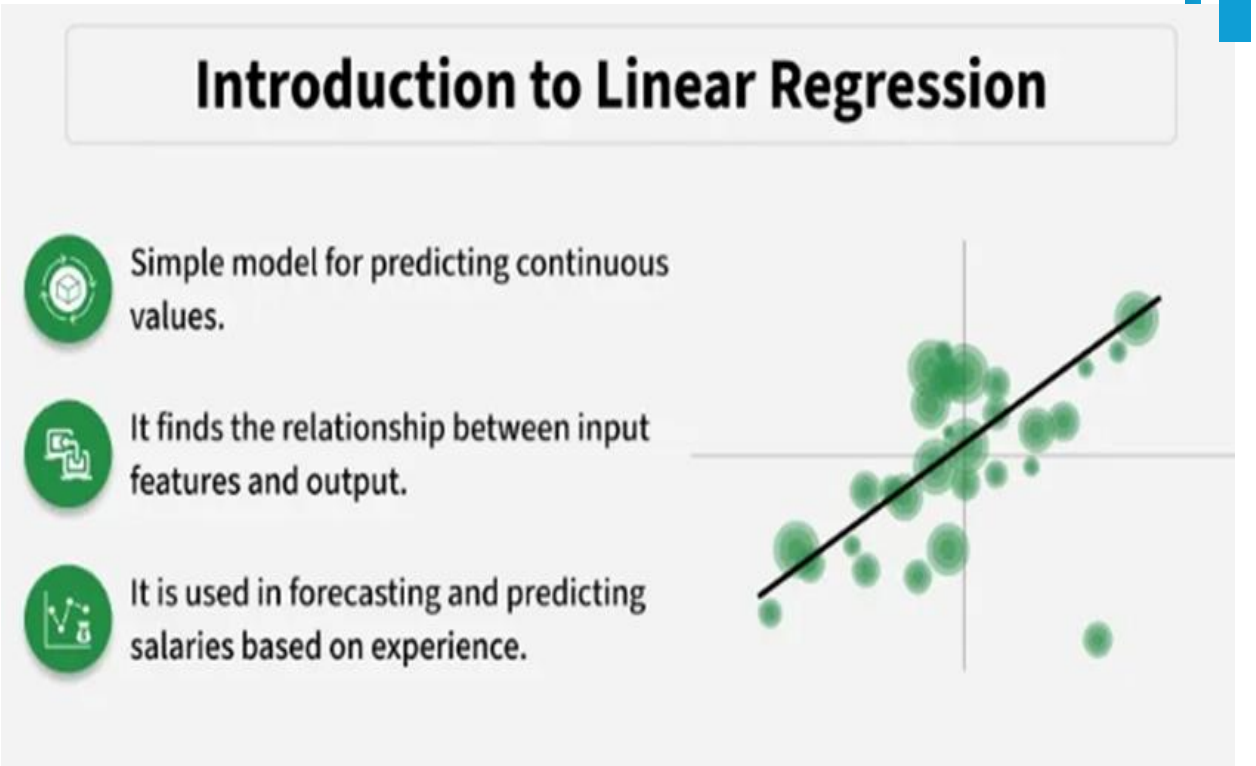
Linear Regression

- We want to predict a student's exam score based on the number of hours they studied. We observe that as study hours increase, exam scores also tend to increase.
- In this example:
- **Independent variable (input):** Hours studied, because it is the factor we observe or control.
- **Dependent variable (output):** Exam score, because it depends on how many hours the student studied.
- The independent variable is used to predict the dependent variable

Linear regression

- **Linear regression** is a machine learning technique used to model and predict a numerical outcome by finding a linear relationship between variables.
- In linear regression:
- We use one or more **independent variables (inputs)** to predict a **dependent variable (output)**.
- The model assumes that the relationship between the input(s) and output can be represented by a straight line.

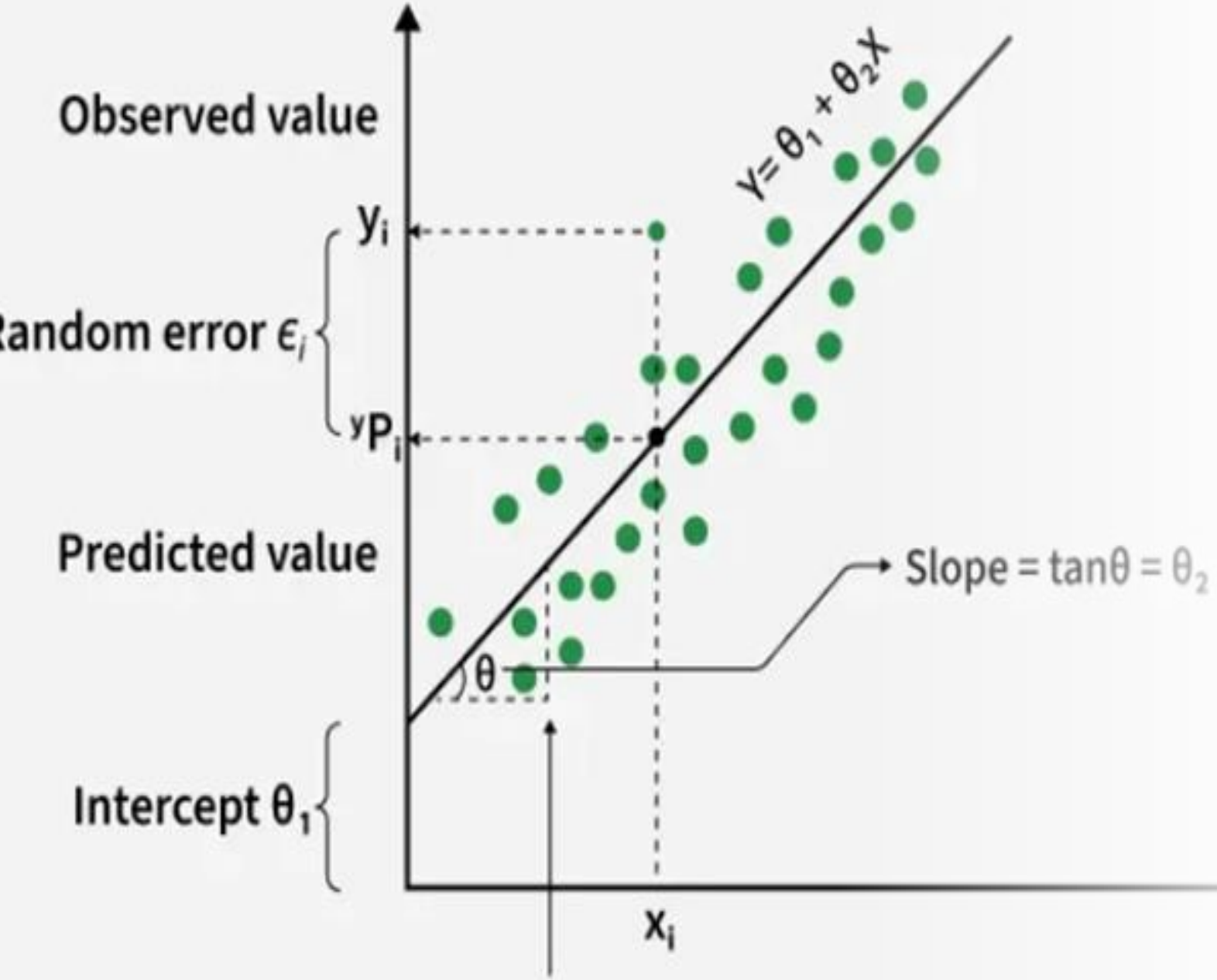


Equation

- The basic idea is expressed by equation
- $Y=mx+b$
- Where:
- y is the predicted output (exam score),
- x is the input (hours studied),
- m is the slope (how much the score changes with each additional hour),
- b is the intercept (the score when study hours are zero).

Best Fit Line in LR

- The **best-fit line** in linear regression is the straight line that most accurately represents the relationship between the independent variable and the dependent variable.
- In the example of **hours studied vs. exam score**, the best-fit line is the line that:
- Comes as close as possible to all the data points on the graph
- Minimizes the overall prediction error between the actual exam scores and the predicted scores
- More specifically, linear regression finds the line that **minimizes the sum of the squared differences (errors)** between the actual values and the values predicted by the line. This method is called **least squares**.
- So, the best-fit line shows the general trend of how exam scores change as study hours increase and is used to make predictions for new data.



Representing best line fit

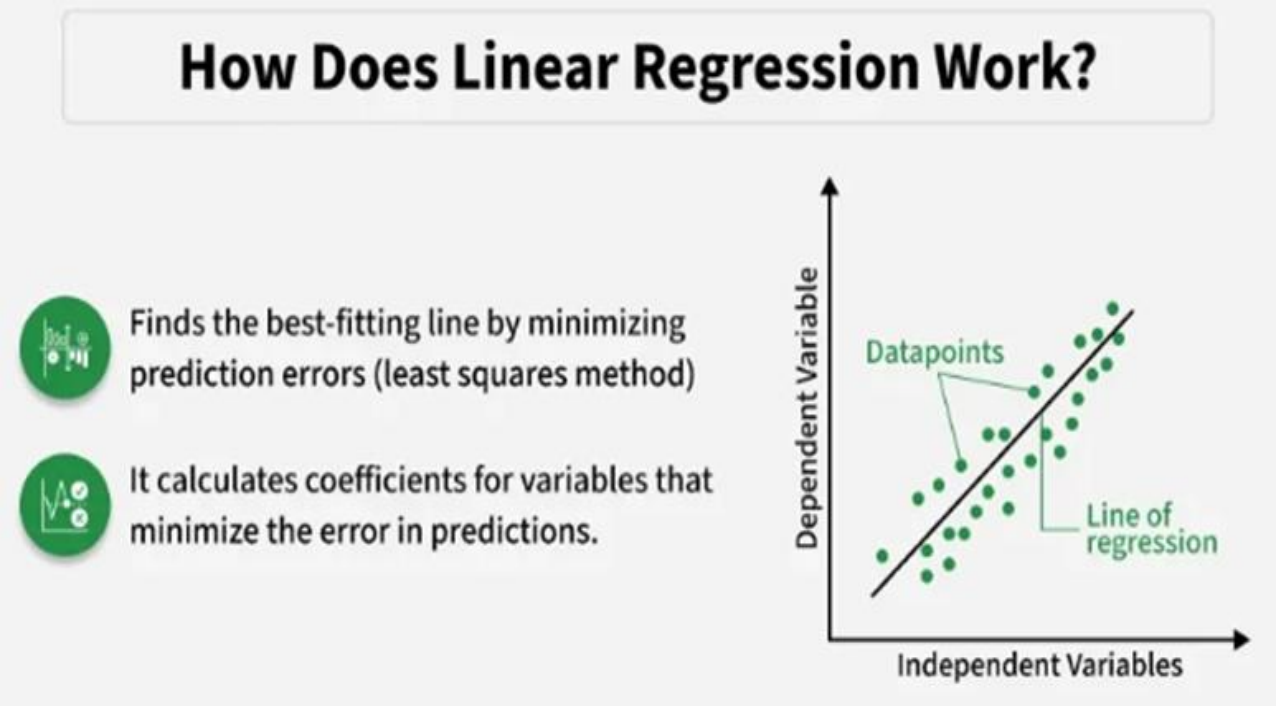
- Here, **Y** is called the **dependent** or **target variable**, and **X** is the **independent variable**, also known as the **predictor** of Y.
  - θ_1 represents the **intercept**, which is the value of Y when X = 0.
  - θ_2 represents the **slope**, which indicates how much Y changes for each one-unit increase in X.
- There are many types of functions or models that can be used for regression, but a **linear function** is the simplest. In this case, **X** can be a single feature or multiple features that represent the problem.

Minimizing the error: The least squares method

- The **least squares method** is a technique used in linear regression to find the **best-fit line** for the data.
- It works by minimizing the **sum of the squared errors** between the actual values and the values predicted by the model.
  An **error** is the difference between an observed value of the dependent variable and the value predicted by the regression line.
- In other words, the least squares method chooses the values of the model parameters (such as the intercept and slope) so that the total of the squared differences between actual and predicted outputs is as small as possible. This ensures that the line fits the data as closely as possible overall.

Interpretation of the Best-Fit Line

- **Slope (m):** The slope shows how much the dependent variable (y) changes for a one-unit increase in the independent variable (x). For example, if the slope is 5, it means that y increases by 5 units for every 1-unit increase in x.
- **Intercept (b):** The intercept is the predicted value of y when x = 0. It represents the point where the line crosses the y-axis.
- In linear regression, certain assumptions are made to ensure the reliability of the model's results.



Limitations:

Assumes Linearity: Linear regression assumes a linear relationship between variables. If the actual relationship is non-linear, the model may produce inaccurate results.

Sensitivity to Outliers: Outliers can have a strong influence on the slope and intercept, potentially distorting the best-fit line.

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

Where:

- $\hat{y}$ is the **predicted value** of the dependent variable
- $x_1, x_2, \ldots, x_n$ are the **independent variables**
- $\theta_1, \theta_2, \ldots, \theta_n$ are the **coefficients (weights)** associated with each predictor
- θ_0 is the **intercept**

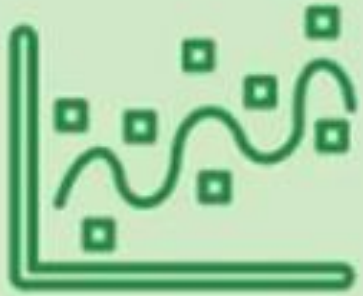
Multiple Linear Regression

- Multiple linear regression is used when there is **one dependent variable** and **two or more independent variables**. It models the relationship between output and multiplr inputs.

Real-World Use Cases of Linear Regression

Stock Market Prediction

Real Estate Price Prediction



Medical Risk Prediction

Sales Forecasting